FIELD GUIDE

HOW A SIGNAL IS DETECTED

An illustrated, source-based walkthrough of the mbioTraders signal engine, from multi-timeframe agreement to entry, stop loss, take profit and browser notification.



05
Trend frames
must align

02
Strong 5m candles
before the entry

01
Third-candle open
sets the entry

Based on the mbioTraders signal detection rules

Diagrams and prices are illustrative, not live trading recommendations.

31 OCT 2026

01 / DECISION MAP

How the engine reaches a signal

The algorithm checks conditions in sequence. A failed gate means WAIT - never a guessed entry.

First: protect any ACTIVE signal
Existing positions are reconciled before news checks. A new setup is evaluated only when a pair has no active signal.

1	**NEWS GUARD** High-impact events: 3h before to 2h after. Missing calendar = WAIT.	WAIT if failed
2	**REAL DATA** Live feed, closed 5m candles, and recent consecutive bars are required.	WAIT if failed
3	**MULTI-DAY BIAS** H1, H4 and D1 must agree over 3/5/7 trading sessions (default 7).	WAIT if failed
4	**FIVE-FRAME ALIGNMENT** 15m, 30m, H1, H4 and D1 must agree; H1 ADX >= 16.	WAIT if failed
5	**COUNTERTREND PULLBACK** At least one opposite-direction 5m candle in the six lead-up bars.	WAIT if failed
6	**TWO STRONG CANDLES** Two consecutive completed 5m candles in the trend direction; body >= 60%.	WAIT if failed
7	**ENTRY + PROTECTION** Actual third 5m open within 90 sec; older confirmed swing; TP = 2R.	WAIT if failed
8	**ACTIVATE + ALERT** No existing or duplicated signal for the pair; notify if Alerts are enabled.	WAIT if failed

All passed = BUY or SELL
Signals are possible only when the live candle and actual opening price are available. Passing a score alone cannot create a signal.

02 / TREND SELECTION

Five charts must agree

The 5-minute chart decides timing only. It cannot overrule the larger-direction consensus.

Example: bullish alignment

The dashboard needs a complete, non-neutral direction across every required timeframe.

TIMEFRAME	DIRECTION	ROLE
15m	BUY / BULLISH	Indicator consensus
30m	BUY / BULLISH	Indicator consensus
1H	BUY / BULLISH	Multi-day trend + ADX
4H	BUY / BULLISH	Multi-day trend
1D	BUY / BULLISH	Multi-day trend
5m	ENTRY ONLY	Entry trigger only

5/5 aligned - proceed to the 5-minute setup

How direction is measured

15m + 30m: a 16-vote indicator mix (EMA, RSI, MACD, ADX, CCI and others) produces BUY, SELL or NEUTRAL.

H1 + H4 + D1: directional price change and slope across completed trading sessions must agree.

Additional strength gates

H1 ADX >= 16

7D DEFAULT

3D and 5D are selectable. Missing daily/session coverage means WAIT.

What if only four align?

A 4/5 outcome is not good enough. The 5m can be bullish during an overall SELL setup, or bearish during a BUY setup - its job is to form a pullback and confirm the next entry, not vote in the trend matrix.

03 / PRECISE CANDLE RULE

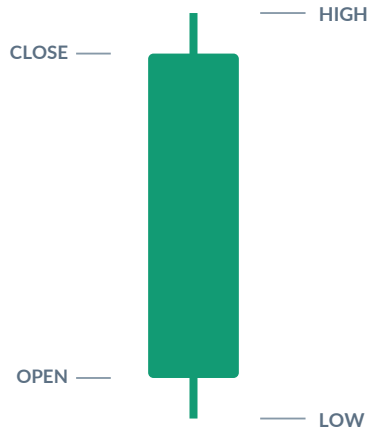
What makes a candle strong?

Only completed candles count. Candle direction must agree with the intended BUY or SELL.

BODY / RANGE RULE

|Close - Open| / (High - Low) >= 0.60

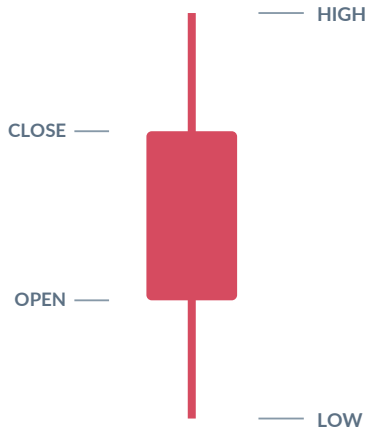
QUALIFIES



80.0%

Body / full candle range

DOES NOT QUALIFY



41.7%

Body / full candle range

The 5-minute entry sequence

1. **PULLBACK** — Against the main trend
2. **CANDLE 1** — Strong trend candle
3. **CANDLE 2** — Another strong candle
4. **CANDLE 3** — Actual OPEN -> entry

Important timing details

The code checks the six completed 5m bars before the two confirmations and looks for at least one countertrend candle. It then needs two consecutive trend-direction candles, each >= 60% body. The third candle must exist in the live feed; its opening price cannot be substituted with the second candle close. Entry is accepted only within 90 seconds of that open.

04 / WORKED BUY CASE

BUY example: from pullback to entry

The plotted candles are synthetic teaching examples. The 5-minute rules and price arithmetic match the trading logic.



The chart shows the exact earlier pivot, two completed confirmations, and a third-candle OPEN marker.

ENTRY	STOP LOSS	RISK / 1R	TAKE PROFIT
2642.5000	2638.0000	4.5000	2651.5000

Why the signal qualifies

1. All five higher frames show BUY and H1 ADX is at least 16.
2. One or more bearish candles appear before the 09:50 + 09:55 bullish confirmations.
3. Their body shares are 82.4% and 78.6%; entry at the actual 10:00 open.
4. A confirmed older 5m swing low is 2638.0000; TP = entry + 2 x risk.

The older pivot is a strict 5-bar local low/high (two candles on each side); never the latest wick.

05 / WORKED SELL CASE

SELL example: from pullback to entry

The plotted candles are synthetic teaching examples. The 5-minute rules and price arithmetic match the trading logic.



ENTRY	STOP LOSS	RISK / 1R	TAKE PROFIT
2628.3000	2633.1000	4.8000	2618.7000

Why the signal qualifies

1. All five higher frames show SELL and H1 ADX is at least 16.
2. One or more bullish candles appear before the 09:50 + 09:55 bearish confirmations.
3. Their body shares are 83.8% and 81.8%; entry at the actual 10:00 open.
4. A confirmed older 5m swing high is 2633.1000; TP = entry - 2 x risk.

The older pivot is a strict 5-bar local low/high (two candles on each side); never the latest wick.

06 / RISK CONTROL

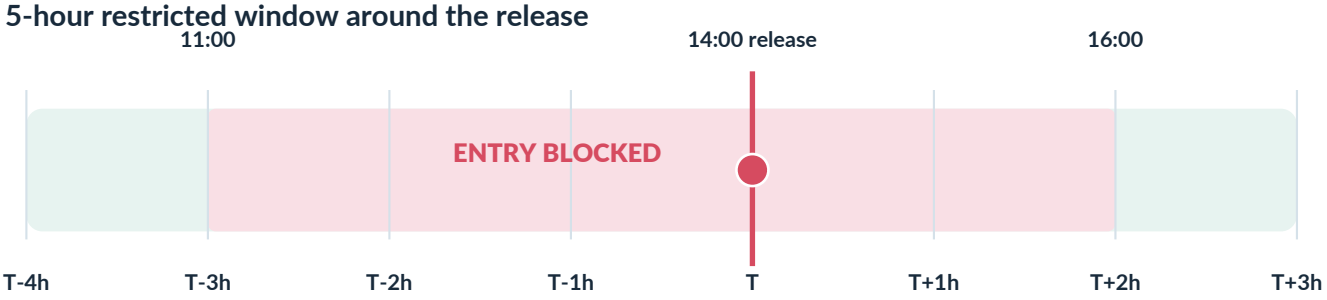
Economic news can pause a setup

The news filter is checked for each market before a new tracked signal is activated.

Example: relevant high-impact release at 14:00 UTC

No new entries are allowed from 11:00 through 16:00 UTC, including the boundary times.

5-hour restricted window around the release



Outside the shaded window, the calendar guard can be clear if no other relevant event overlaps. An existing ACTIVE signal is not cancelled by news.

Which economic releases are relevant?

XAUUSD	United States
EURUSD	United States, Eurozone, Germany
GBPUSD	United States, United Kingdom
USDJPY	United States, Japan
AUDUSD	United States, Australia
USDCAD	United States, Canada

When the calendar cannot be trusted

The feed refreshes roughly every 90 seconds while open; an unreadable or older-than-4-minute snapshot blocks new entries. Missing precise release times produce a wider date-based blackout. Only confirmed high-impact events are considered; surprise news may still occur.

07 / AFTER DETECTION

What happens after activation?

A detected entry creates a tracked paper signal, then optionally a real system browser notification.

ACTIVE SIGNAL LIFECYCLE

NEW SIGNAL
one active per pair

VERIFY 5m BARS
completed candles only

TAKE PROFIT

STOP LOSS

If both TP and SL are touched inside one completed 5m candle, the code records SL (conservative). A later trend flip alone never cancels an active signal.

REAL BROWSER NOTIFICATION



Prerequisites: Alerts On, browser permission GRANTED, HTTPS or localhost, a supported browser, and the dashboard running. Duplicate notifications are suppressed by signal ID.

Persistence & history

Signals are saved in browser localStorage and survive refreshes. The system allows one active signal per pair; a new setup never overwrites it.

100 completed records

2 results per page

Browser-closed limitation

Notifications can appear while the dashboard tab is unfocused, depending on browser/OS behavior. This dashboard does NOT include server-side push, a 24/7 monitoring worker or notifications after closing all tabs.

NO BACKGROUND SERVER

08 / INTERPRETING THE RESULT

A confidence score is not a win rate

Every hard gate must pass. A high displayed score alone is not permission to enter.

100-point internal setup score



Frame agreement	60 pts	H1 ADX strength	10 pts
5m pullback	10 pts	Two strong candles	15 pts
Consecutive 5m data	5 pts		

Agreement contributes up to 60 points (proportional to the number aligned). The other terms add 10 + 10 + 15 + 5 when satisfied. This is not a calibrated probability of profit.

Why the dashboard may show WAIT

- **Calendar missing or event near**
  High-impact news blocks entries T-3h through T+2h.
- **Incomplete higher timeframe data**
  Not enough completed 3D/5D/7D sessions.
- **Timeframes mixed / neutral**
  Not all five required frames agree on BUY or SELL.
- **Weak H1 trend strength**
  H1 ADX is below 16.
- **Stale or gapped 5m candles**
  No recent continuous real OHLC sequence.
- **No valid pullback or 2 candles**
  Opposing candle missing, or body ratio below 60%.
- **Third candle unavailable or late**
  Actual open absent, or more than 90s have elapsed.
- **Swing / lifecycle gate failed**
  Older pivot missing, risk invalid, or pair already active.

A practical reminder

This is a signal-only dashboard, not an order execution system. SL/TP levels are calculated from historical candles; they do not guarantee execution price or a profitable result. Data delays, calendar changes, spread and slippage can affect real-world outcomes.

09 / IMPLEMENTATION DETAILS

Exact checks and source reference

A compact technical audit sheet: how to recognize a valid candidate and trace each rule to its implementation.

Production rules at a glance

1 **NEWS / FRESHNESS**

Calendar <= 4 min old; high-impact block: T-3h to T+2h. Last completed 5m bar <= 15 min old; last six closed bars consecutive; entry pattern checks ten consecutive bars.

2 **MOMENTUM / ALIGNMENT**

3, 5 or 7 completed daily sessions (7 default); H1, H4, D1 trends agree; 15m + 30m indicator votes agree; H1 ADX >= 16.

3 **CANDLE PATTERN**

At least one countertrend 5m candle in six bars preceding the confirmations. Two successive completed trend candles: |C-O|/(H-L) >= 0.60 each.

4 **ENTRY / STOP / TARGET**

Use the next actual 5m candle OPEN within 90s. Swing confirmed with 2 candles on either side; SL = swing wick; positive risk required; TP = entry +/- 2 x risk.

5 **LIFECYCLE / ALERT**

One active signal per pair. Reconcile on completed 5m bars; stop prioritized if both touched. Notify on new active signal if Alerts On + permission granted.

LOGIC SUMMARY (SIMPLIFIED)

IF an active signal exists: reconcile it; do not replace it.

ELSE IF news invalid / live feed invalid: WAIT.

ELSE IF momentum incomplete OR required alignment < 5/5: WAIT.

ELSE IF H1 ADX < 16 OR 5m pullback missing: WAIT.

ELSE IF either closed confirmation body < 60%: WAIT.

ELSE IF actual third OPEN missing OR later than 90s: WAIT.

ELSE IF earlier confirmed swing missing / risk <= 0: WAIT.

ELSE activate BUY or SELL; set SL at pivot, TP at 2R.

Primary source files: Signal detection, momentum evaluation, economic calendar, signal ledger, and browser notification modules.